

P2 : Urban Heat Island Analysis using Land Surface Temperature and
Built-up Density

**Project Report Submitted in fulfillment of the Requirements for
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Geospatial Intelligence

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Project 2 : Urban Heat Island Analysis using Land Surface Temperature and Built-up Density

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Objective

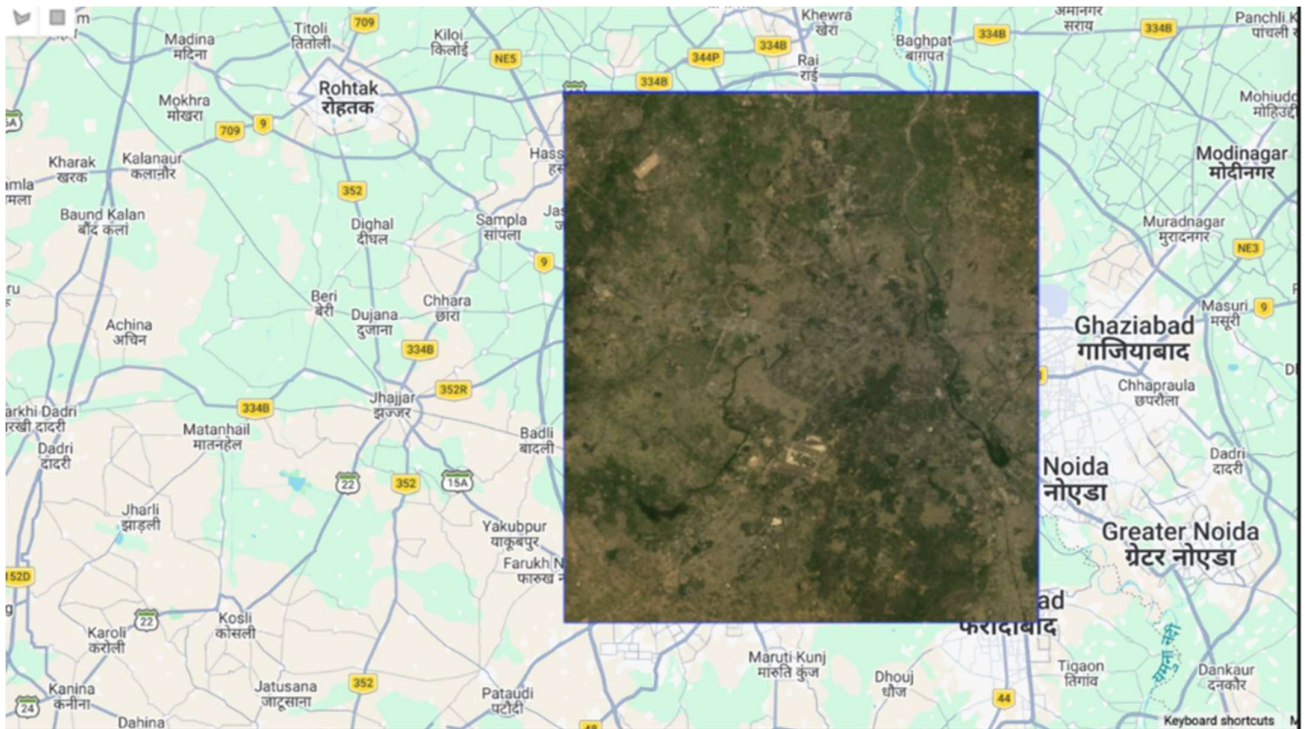
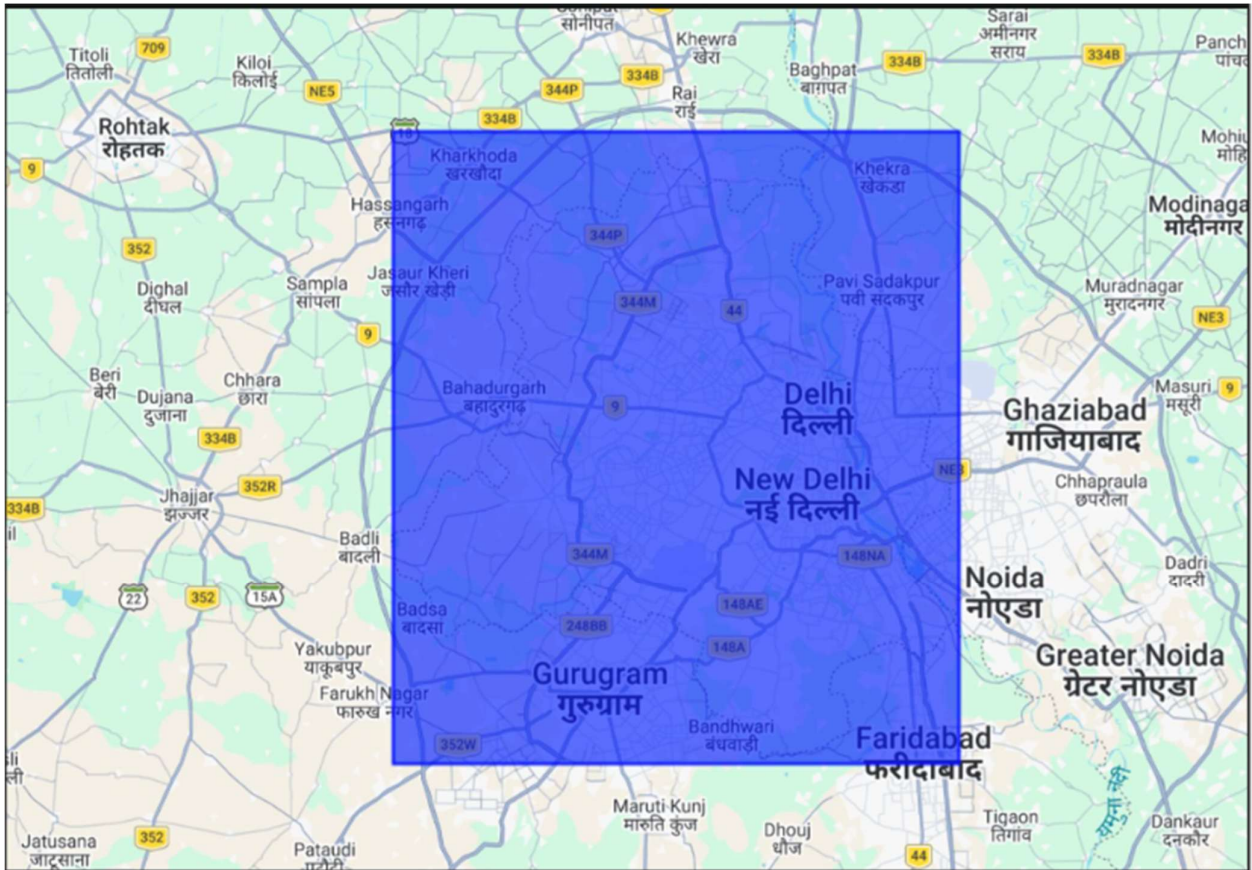
The main objective of this project is to study the Urban Heat Island (UHI) effect in Delhi by deriving Land Surface Temperature (LST) from satellite data and analyzing its relationship with built-up areas. The aim is to identify high-temperature zones and understand how urbanization influences temperature variation across the city.

Study Area

The study area selected for this project is Delhi (National Capital Territory of India), which is one of the most densely populated and rapidly urbanizing regions in the country.

Delhi experiences significant temperature variations due to increasing built-up areas, reduced vegetation cover, and high human activity. These factors make it an ideal region for analyzing the Urban Heat Island effect.

The boundary of Delhi was used as the Area of Interest (AOI), and all satellite data was clipped according to this boundary for accurate analysis.



Data Used

Satellite Data:

- Landsat-8 OLI/TIRS (for LST calculation)
 - Sentinel-2

Required Bands (Landsat-8):

- Band 10 (Thermal Infrared – TIRS)
 - Band 4 (Red)
 - Band 5 (NIR)

Ancillary Data:

- Area of Interest (AOI) shapefile
- Optional: Land use/land cover or urban boundary data

Methodology

1) cloud-free Landsat-8 satellite imagery for Delhi was collected. The required bands were loaded and clipped to the study area.

The thermal band (Band 10) was converted from Digital Number (DN) to Top of Atmosphere (TOA) radiance using raster calculations. After that, radiance values were converted into brightness temperature in Kelvin.

2) NDVI was calculated using Red and NIR bands to estimate vegetation cover. Based on NDVI values, land surface emissivity was calculated.

Using emissivity correction, the brightness temperature was converted into Land Surface Temperature (LST) in degree Celsius.

3) For built-up analysis, urban areas were identified using spectral characteristics and classification

techniques. Areas with dense construction were marked as high built-up zones.

The LST values were then classified into different temperature categories such as low, moderate, and high temperature zones. High temperature areas were identified as Urban Heat Island (UHI) zones.

Finally, spatial analysis was performed to compare built-up density with temperature distribution, and raster outputs were converted into vector format to calculate area statistics.

QGIS-Based Land Surface Temperature (LST) and Urban Heat Island (UHI) Workflow

The Landsat-8 satellite imagery was imported into QGIS for processing. Band 4, Band 5, and Band 10 were clipped to the selected Area of Interest. The clipped thermal band was converted from Digital Number to Top of Atmosphere Radiance using the radiometric rescaling factors available in the Landsat metadata.

Step 1 : DN to TOA Radiance

$$L_{\lambda} = M_L \times Q_{cal} + A_L$$

where:

L_{λ} = TOA Radiance

M_L = Radiance multiplicative scaling factor

A_L = Radiance additive scaling factor

Q_{cal} = Digital Number (DN)

Step 2 : Brightness Temperature

$$BT = \frac{K_2}{\ln \left(\frac{K_1}{L_\lambda} + 1 \right)}$$

where:

BT = Brightness Temperature (Kelvin)

K 1,K 2 = Thermal calibration constants from Landsat metadata

Step 3 : NDVI Calculation

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

where:

NIR = Band 5

Red = Band 4

Step 4 : Proportion of Vegetation

$$P_v = \left(\frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \right)^2$$

Step 5 : Land Surface Emissivity

$$\varepsilon = 0.004 \times P_v + 0.986$$

Step 6 : Land Surface Temperature

$$LST = \frac{BT}{1 + \left(\frac{\lambda \times BT}{\rho} \right) \ln(\varepsilon)} - 273.15$$

where:

- $\lambda = 10.895 \mu m$
- $\rho = 1.438 \times 10^{-2} m \cdot K$
- ε = Land Surface Emissivity

Step 7: Built-up Area Analysis

The Normalized Difference Built-up Index (NDBI) was computed to identify built-up regions.

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR}$$

where:

SWIR = Band 6

NIR = Band 5

Positive NDBI values generally indicate built-up areas, while negative values represent vegetation or water bodies.

Step 8: Urban Heat Island Identification

Temperature Range	Category
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Low	Cool Zone
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Moderate	Normal Zone
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High	Urban Heat Island
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Step 9: Spatial Analysis

The spatial relationship between LST and built-up density was analyzed by overlaying the LST and NDBI layers. Regions with dense built-up surfaces generally corresponded to higher surface temperatures, indicating a positive correlation between urbanization and heat intensity.

Step 10: Area Estimation

The classified UHI raster was converted into vector polygons using the Polygonize tool, and the area of each hotspot was calculated using the field calculator to quantify the spatial extent of Urban Heat Island zones.

Google Earth Engine (GEE) Workflow (Urban Heat Analysis):

Area of interest

```
var aoi = ee.Geometry.Rectangle([76.84, 28.40, 77.35, 28.90]);

Map.centerObject(aoi, 10);
Map.addLayer(aoi, {color: 'blue'}, 'Delhi Boundary');

var collection = ee.ImageCollection('LANDSAT/LC08/C02/T1_L2')
  .filterBounds(aoi)
  .filterDate('2023-01-01', '2023-12-31')
  .filter(ee.Filter.lt('CLOUD_COVER', 40));

print('Number of Images:', collection.size());

var image = collection.median().clip(aoi);

print('Available Bands:', image.bandNames());
```

Rgb

```
var sr = image.select([
  'SR_B2',
  'SR_B3',
  'SR_B4',
  'SR_B5',
  'SR_B6',
  'SR_B7'
]).multiply(0.0000275)
  .add(-0.2);

Map.addLayer(
  sr,
  {
    bands: ['SR_B4', 'SR_B3', 'SR_B2'],
    min: 0,
    max: 0.3
  },
  'RGB'
);
```

Ndvi

```
var ndvi = sr.normalizedDifference(['SR_B5', 'SR_B4']).rename('NDVI');

Map.addLayer(
  ndvi,
  {
    min: -0.2,
    max: 1,
    palette: ['white', 'yellow', 'green']
  },
  'NDVI'
);
```

Ndbi

```
var ndbi = sr.normalizedDifference(['SR_B6', 'SR_B5']).rename('NDBI');  
Map.addLayer([  
  ndbi,  
  {  
    min: -0.5,  
    max: 0.5,  
    palette: ['green', 'white', 'red']  
  },  
],  
'NDBI'  
);
```

Lst

```
var lst = image.select('ST_B10')  
  .multiply(0.00341802)  
  .add(149.0)  
  .subtract(273.15)  
  .rename('LST');  
Map.addLayer(  
  lst,  
  {  
    min: 20,  
    max: 50,  
    palette: ['blue', 'cyan', 'green', 'yellow', 'orange', 'red']  
  },  
  'Land Surface Temperature'  
);
```

Uhi

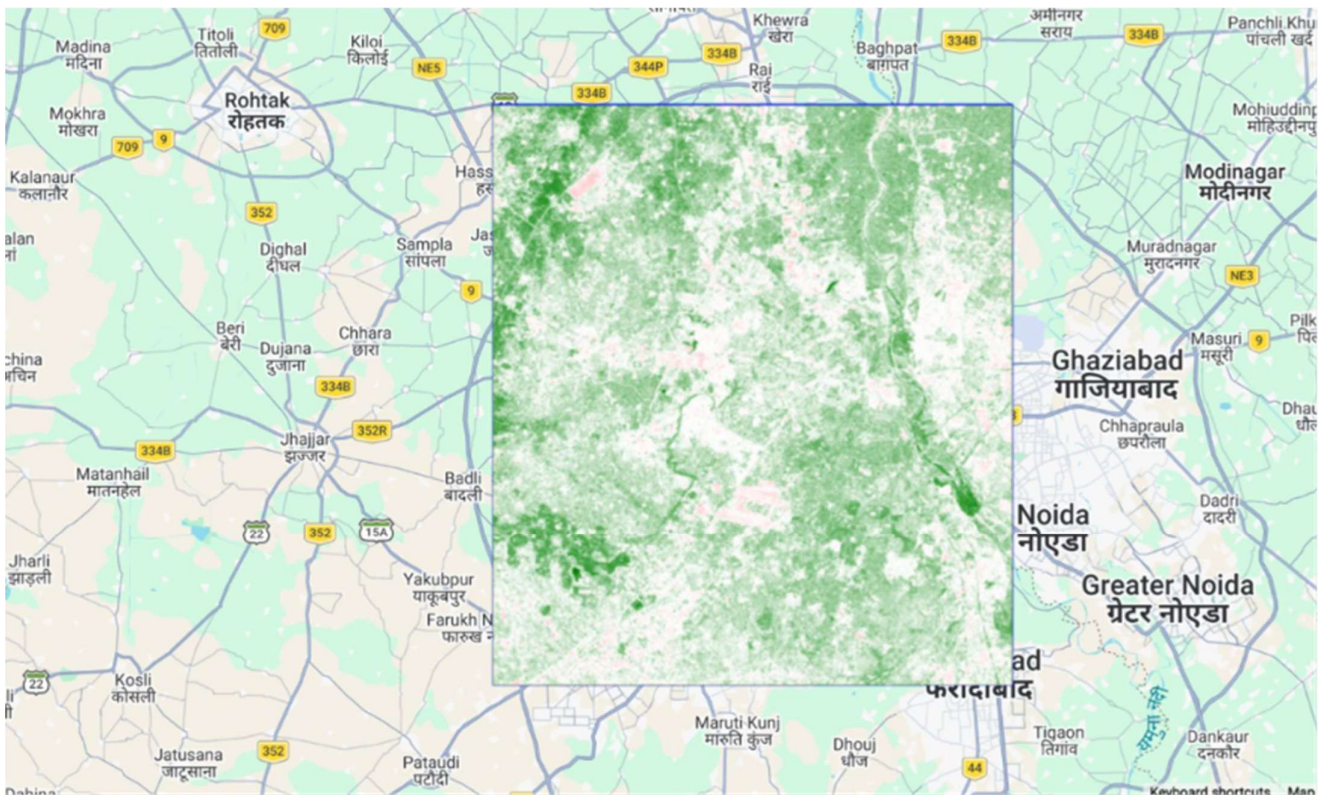
```
var uhi = lst.gt(35);  
Map.addLayer(  
  uhi.updateMask(uhi),  
  {  
    palette: ['red']  
  },  
  'Urban Heat Island'  
);
```

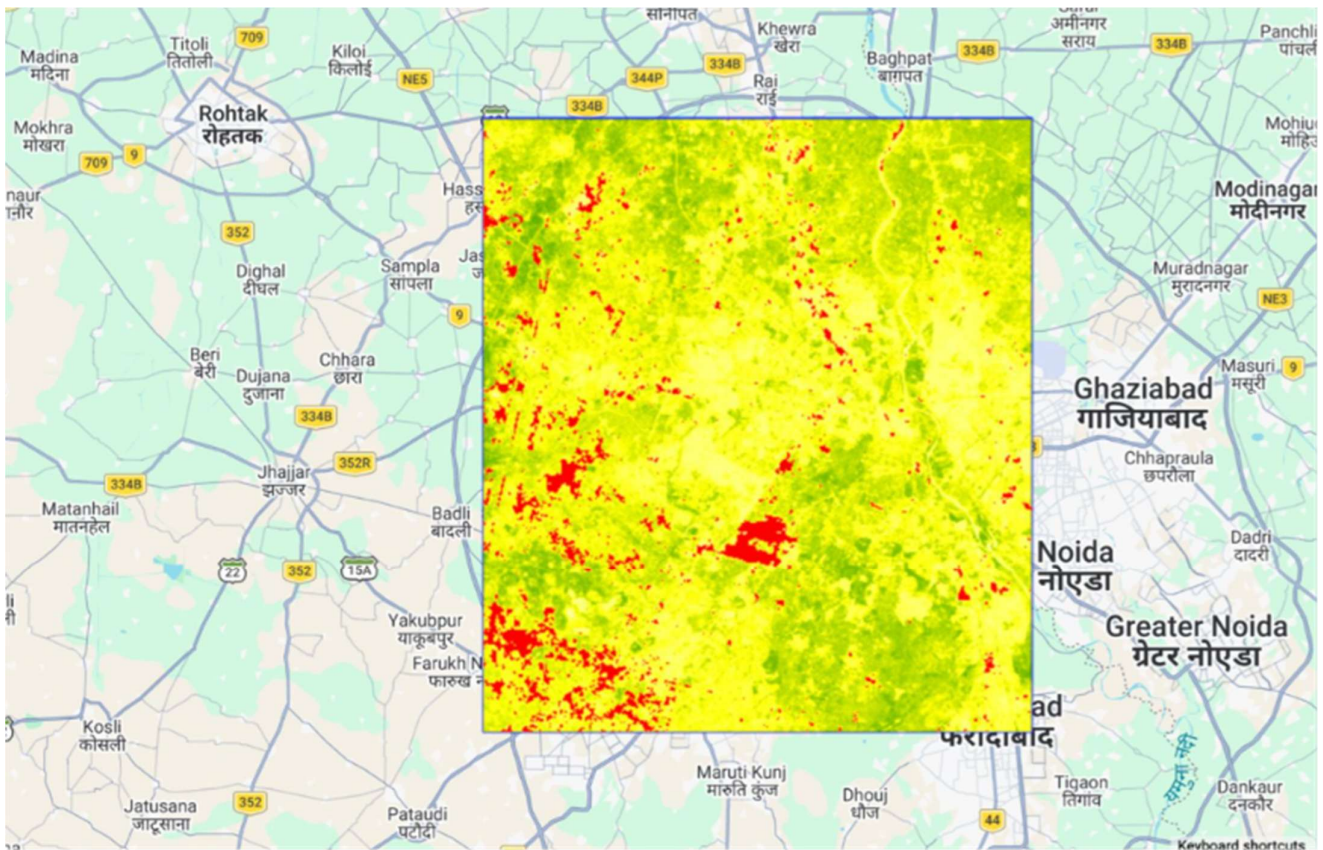
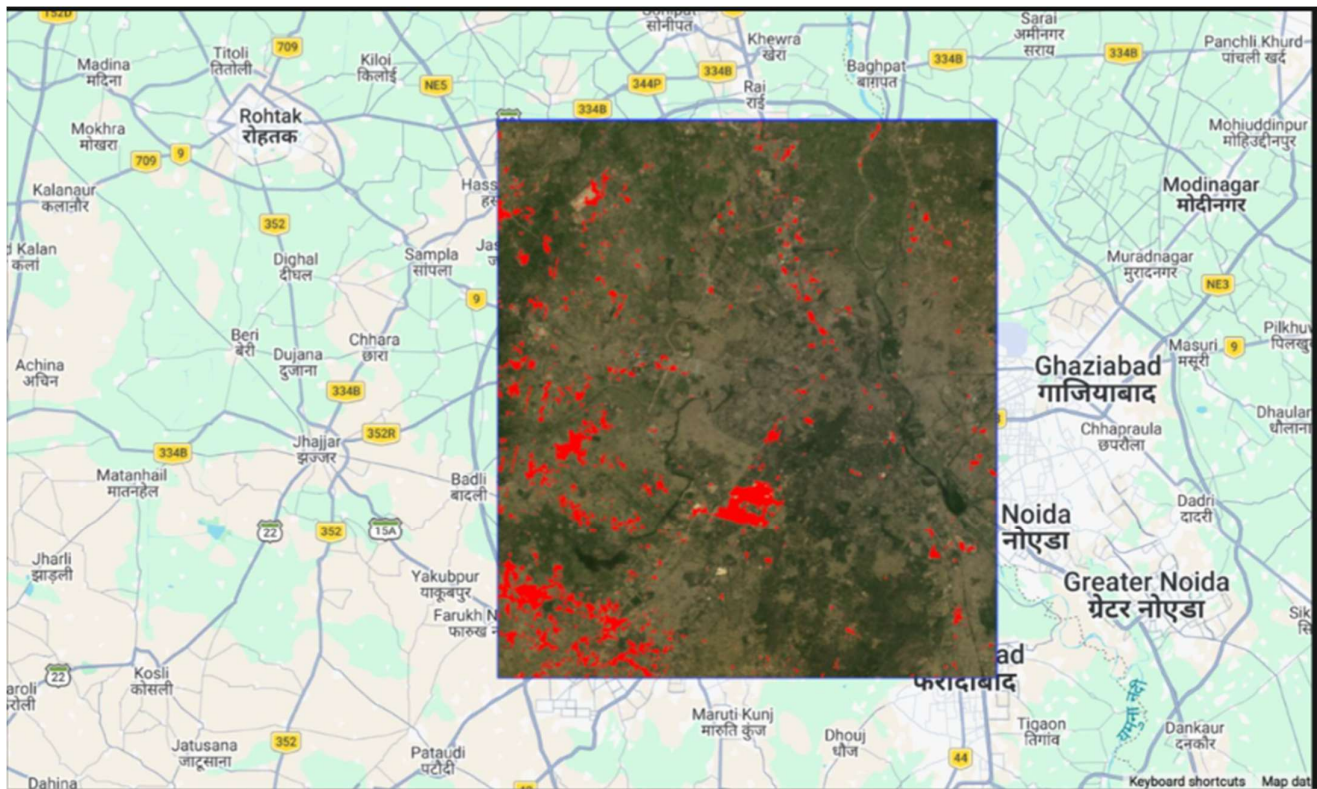
Average temperature

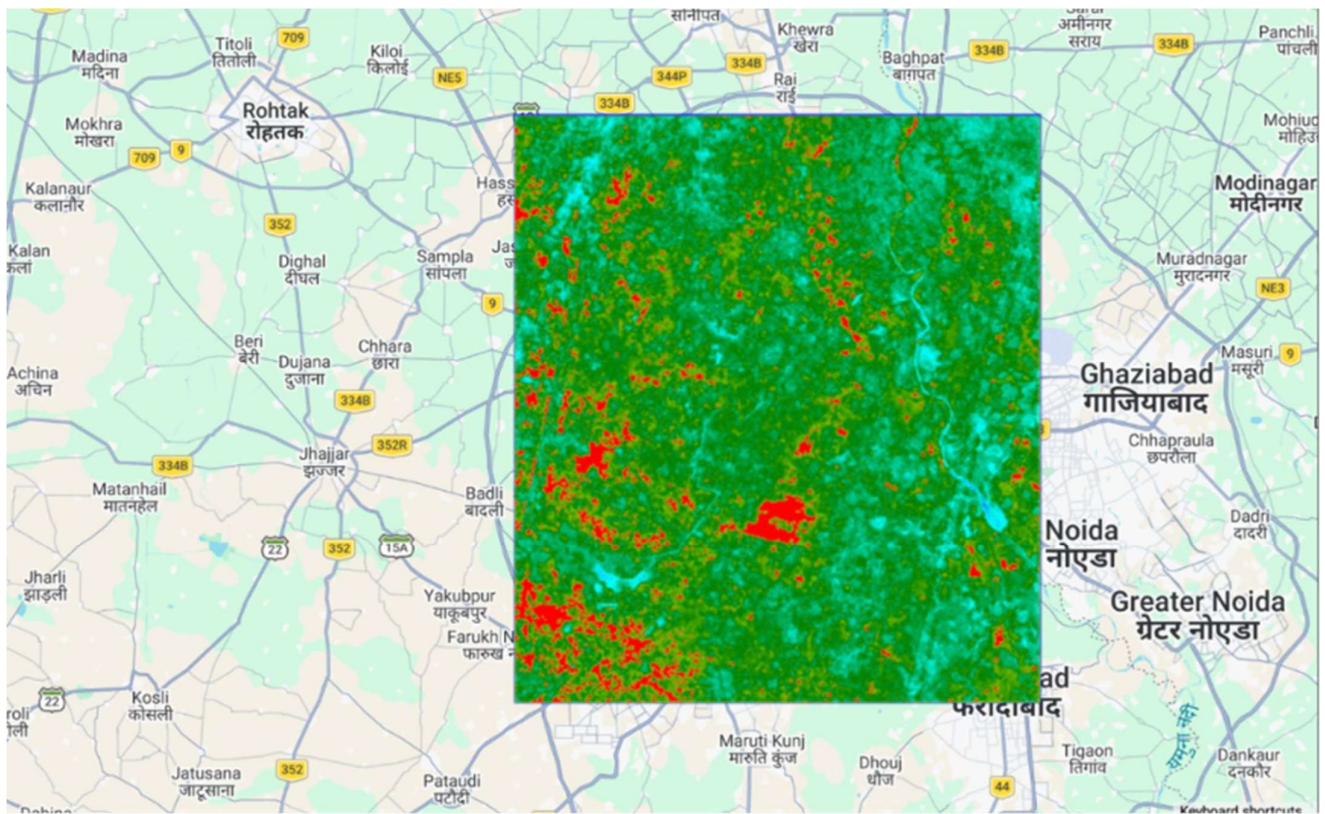
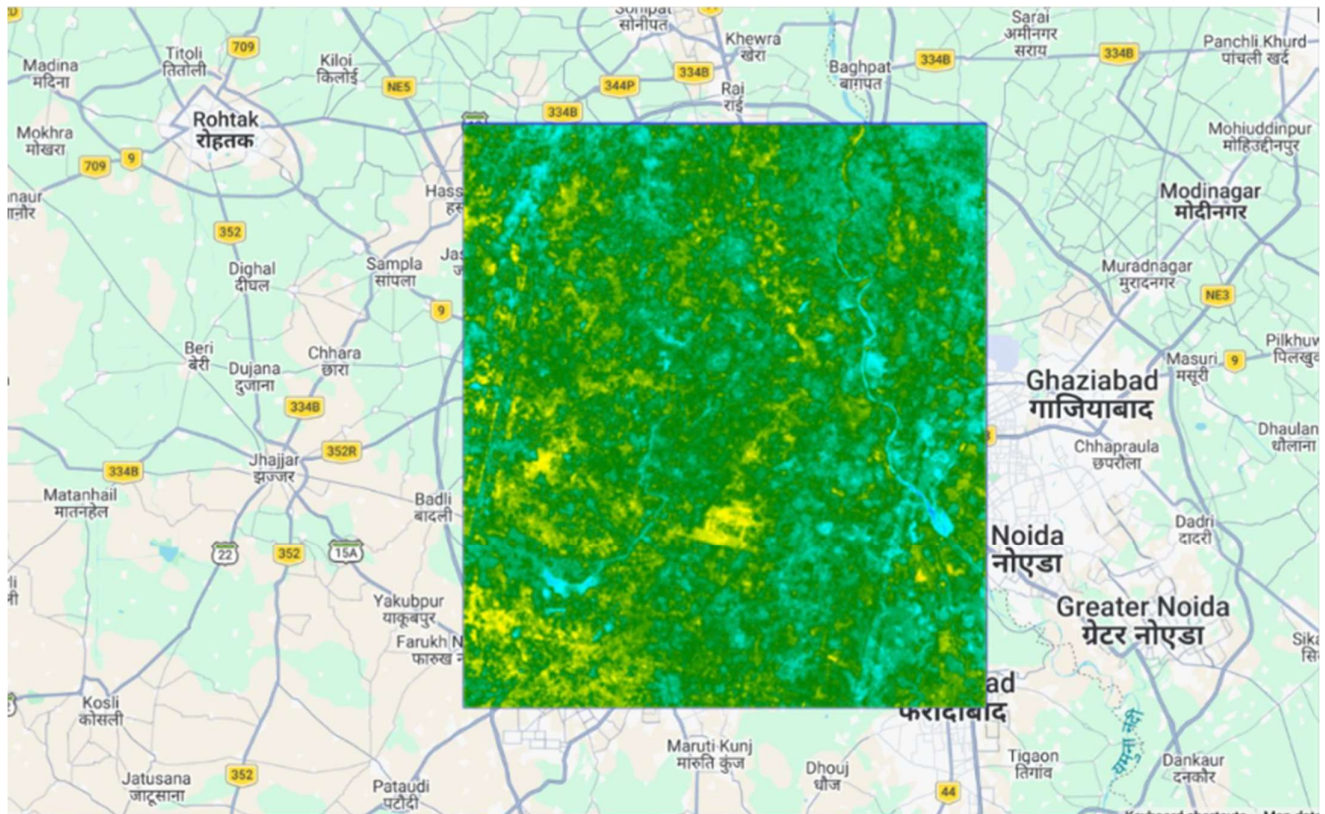
```
var stats = lst.reduceRegion({  
  reducer: ee.Reducer.mean(),  
  geometry: aoi,  
  scale: 30,  
  maxPixels: 1e13  
});  
print('Average Temperature (°C):', stats);
```

Export

```
Export.image.toDrive({  
  image: uhi,  
  description: 'Delhi_UHI_2023',  
  folder: 'EarthEngine',  
  fileNamePrefix: 'Delhi_UHI_2023',  
  region: aoi,  
  scale: 30,  
  maxPixels: 1e13  
});
```







Results

1. Land Surface Temperature (LST) Map

The Land Surface Temperature (LST) map of Delhi revealed spatial variation in surface temperatures. The highest temperatures were observed in densely urbanized regions such as Central Delhi, East Delhi, and major industrial and commercial areas. Lower temperatures were observed near the Yamuna River, urban parks, and vegetated regions.

2. Urban Heat Island (UHI) Zones

Urban Heat Island (UHI) hotspots were identified in highly built-up areas with dense infrastructure and limited vegetation cover. Residential colonies, industrial estates, and commercial districts exhibited significantly higher surface temperatures than surrounding green spaces.

3. Built-up Area and Temperature Relationship

The NDBI analysis indicated a strong positive relationship between built-up density and land surface temperature. Areas with higher NDBI values have higher LST values, while regions with higher NDVI values showed lower temperatures due to the vegetation.

4. Area Statistics of Heat Zones

The study area was classified into low, moderate, and high temperature zones. High-temperature zones were mainly concentrated in densely developed parts of the city, while low-temperature zones were associated with forests, parks, water bodies, and agricultural land.

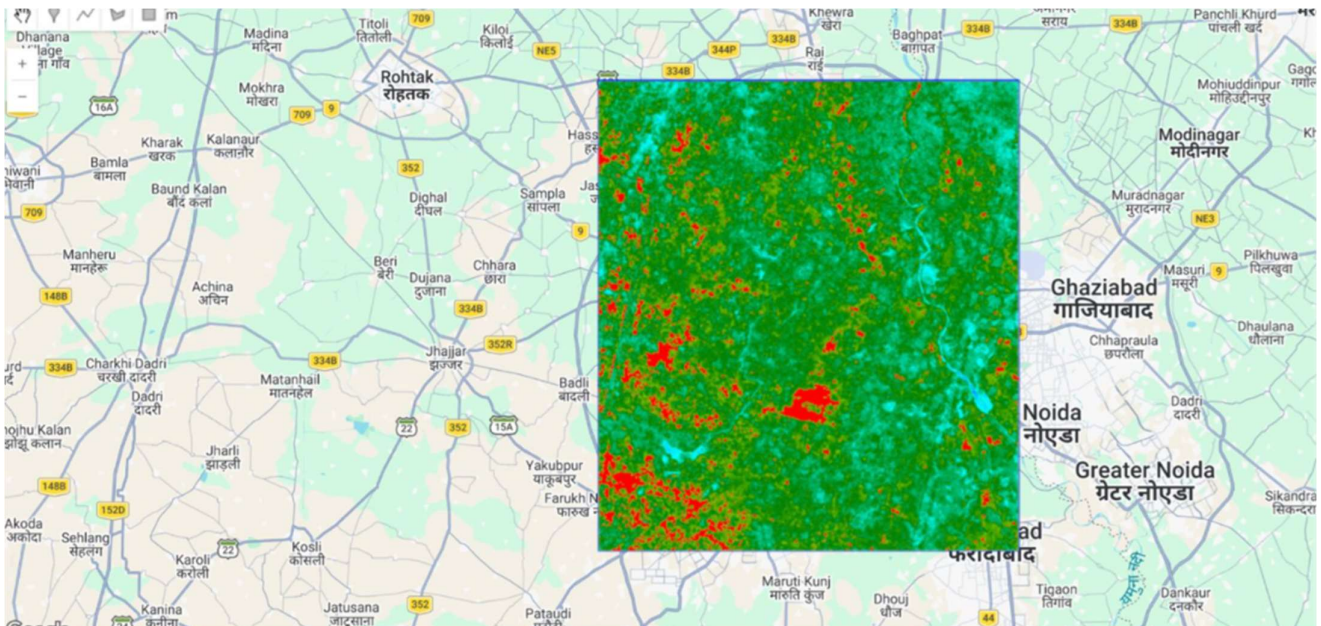
5. Final Cartographic Outputs

The following thematic maps were generated:

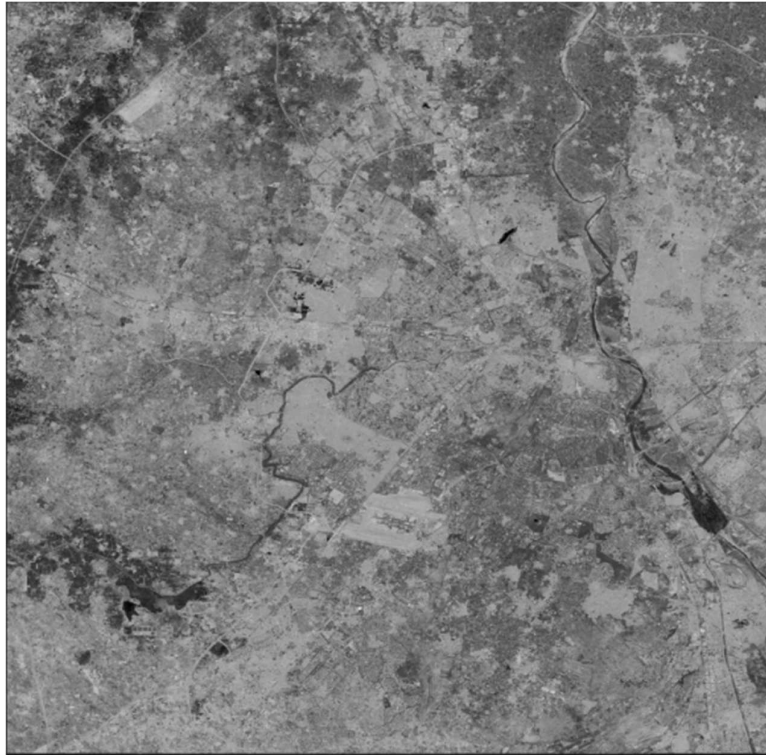
- RGB Satellite Image
- NDVI Map
- NDBI Map
- Land Surface Temperature (LST) Map
- Urban Heat Island (UHI) Map

The results demonstrate that increasing urbanization and reduction in vegetation contribute significantly to elevated land surface temperatures across Delhi.

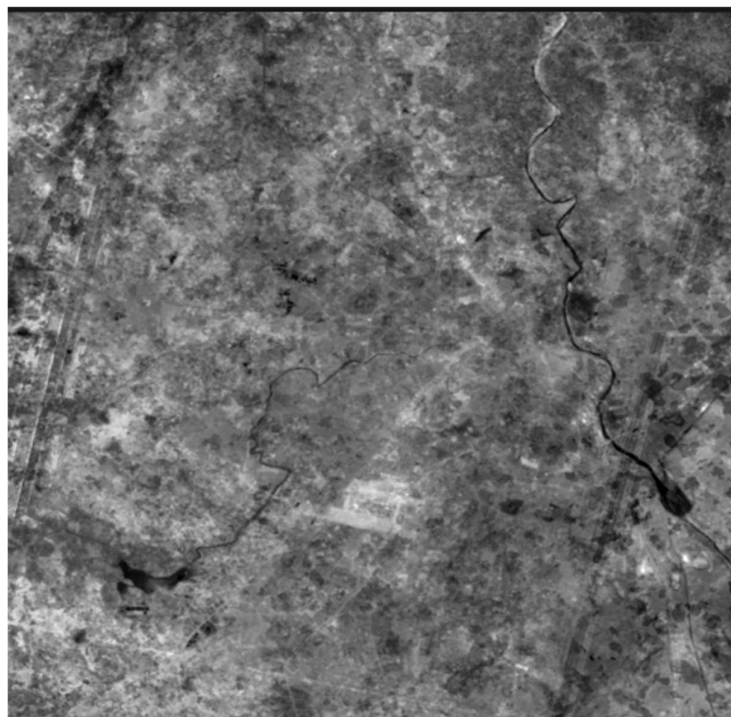
Final map with all layers :



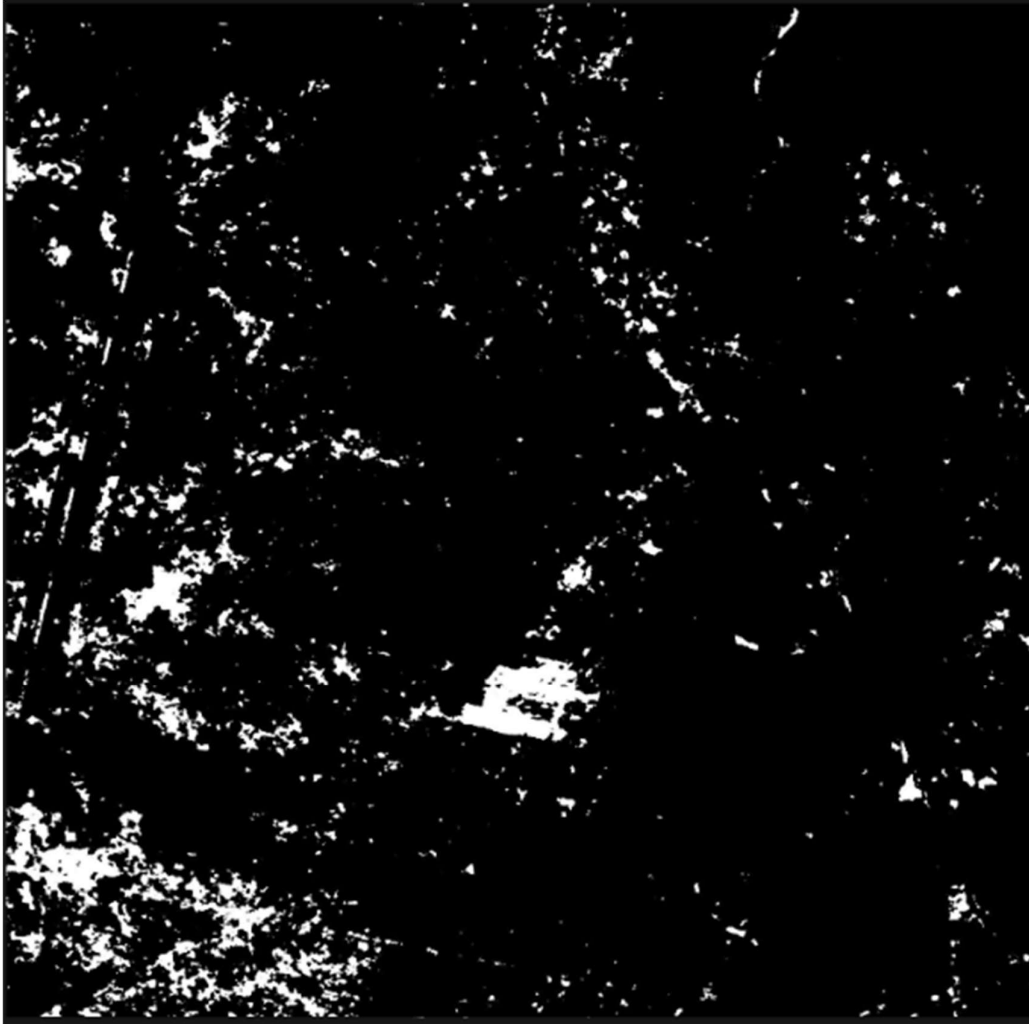
ndbi map :



ndvi map :



urban heat island map :



1st map :



Conclusion

The Urban Heat Island (UHI) analysis of Delhi using Landsat-8 imagery and Land Surface Temperature (LST) demonstrated significant variations in surface temperature across the city.

Areas with dense urban development exhibited higher temperatures, whereas regions with abundant vegetation, parks, and the Yamuna River recorded comparatively lower temperatures. The combined analysis of LST, NDVI, and NDBI highlighted a clear relationship between urbanization and increased temperatures.

The study illustrates the importance of thermal remote sensing and geospatial analysis in identifying heat-prone areas and understanding the environmental impacts of rapid urban expansion. Such information can assist urban planners and policymakers in implementing strategies such as increasing green spaces and improving land-use planning while adopting sustainable development practices to reduce the Urban Heat Island effect.

Although the estimated Land Surface Temperature is influenced by factors such as atmospheric conditions, the integration of vegetation and built-up values provides a comprehensive understanding of Delhi's urban thermal environment. Overall, this project demonstrates the effectiveness of remote sensing and GIS techniques in supporting climate-resilient urban planning and sustainable city development.

References

- Google Earth Engine
- MODIS NDVI Dataset (GEE)
- Google Earth Engine Documentation
- India Space Academy Summer Internship Project Guidelines
- Sentinel-2 Dataset
- NDVI/VCI formulas (class notes)